

Panel MS-AR(1): SOE log real GDP per worker

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Each empirical section has a table of regime AR(1) estimates (standard errors in parentheses) and a plot of country cycles.

1. Model

Joint panel Markov-switching AR(1) around a trend. Three regimes.

$$y_{it} = a_i + g_i t + z_{it}$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}$$

$s_{i,t+1}$ drawn from row s_{it} of Π

Countries are independent given shared Markov parameters; latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1} . σ switches with the regime. Unless noted, ρ is regime-specific and the median μ is pinned at 0 after ordering. Likelihood: Hamilton filter per country, summed.

Trend assumptions in this report

2. Common a and common linear g (joint MLE).
3. Country intercepts a_i , common g (a_i profiled).
4. Country intercepts a_i and slopes g_i (both profiled).
5. Country-specific Christiano-Fitzgerald low-pass trend
(cycle = periods of 2-100 quarters / 25 years), then MS-AR on the cycle.
6. Same CF trend as 5, with every $\mu(s) = 0$.
7. Same CF trend as 5, with one ρ common to all regimes.

Sections 5-7 are two-step (not joint MLE of trend and cycle).

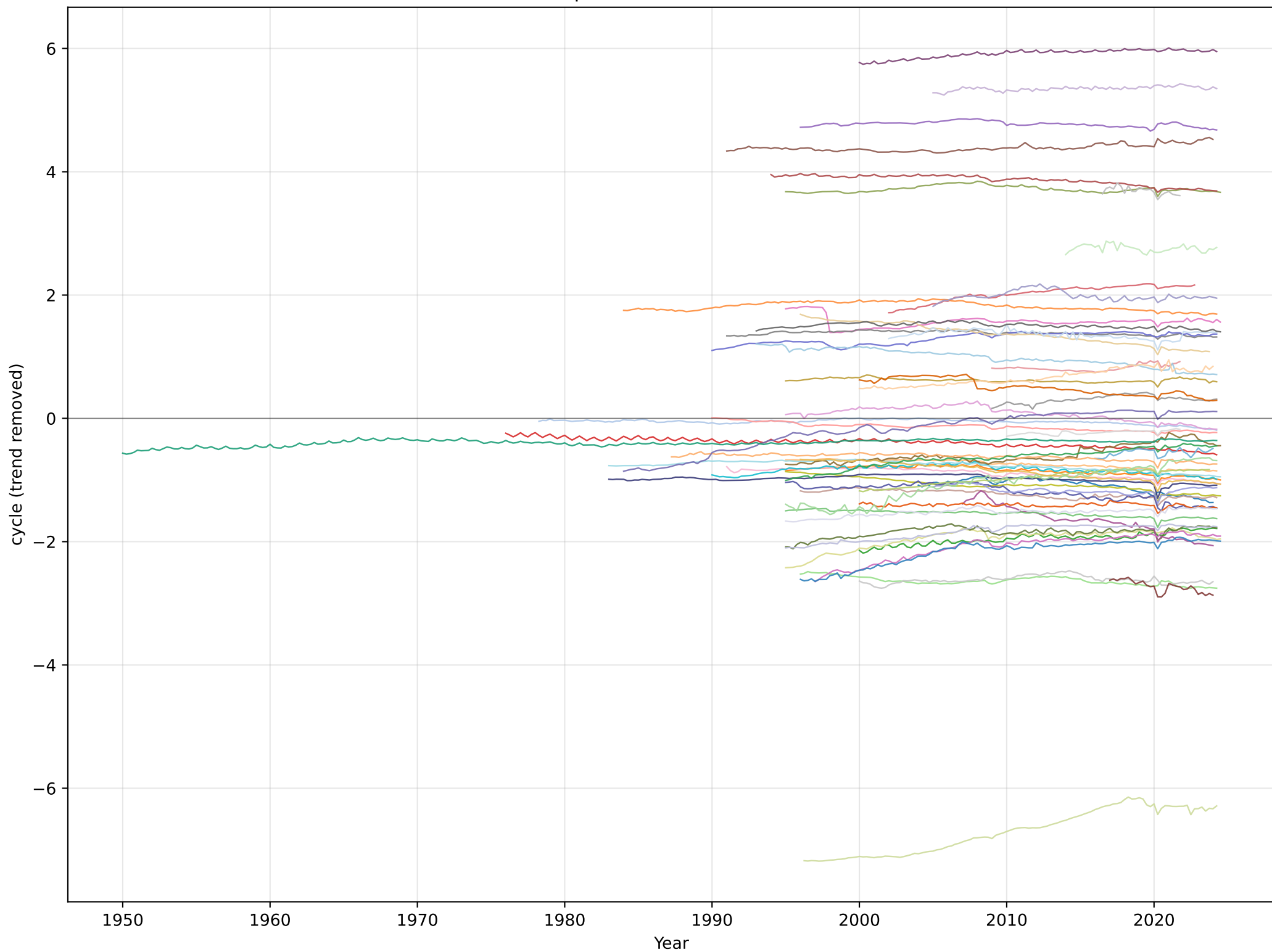
2. Common intercept and linear trend

SOE sample: log real GDP per worker countries in file=62 obs=7004 1950-Q1–2024-Q3
 Regimes: 3 countries: 62 obs: 7004 log-likelihood: 17917.39
 Converged: True CONVERGENCE: RELATIVE REDUCTION OF F <= FACTR*EPSMCH
 a -4.0009 g 0.0139

Regime AR(1) and transition matrix (SEs in parentheses)

	regime 0	regime 1	regime 2
mu	-0.1033	0.0000	0.7480
		(—)	
sigma	0.0099	0.0222	0.0692
	(0.0002)		
rho	0.9900	0.9900	0.9900
Pi from 0	0.9754	0.0070	0.0177
	(0.0044)		(0.0013)
Pi from 1	0.0133	0.9691	0.0176
	(0.0023)	(0.0029)	(0.0021)
Pi from 2	0.0812	0.1445	0.7743
	(0.0068)	(0.0109)	(0.0112)

2. Common intercept and linear trend — detrended series



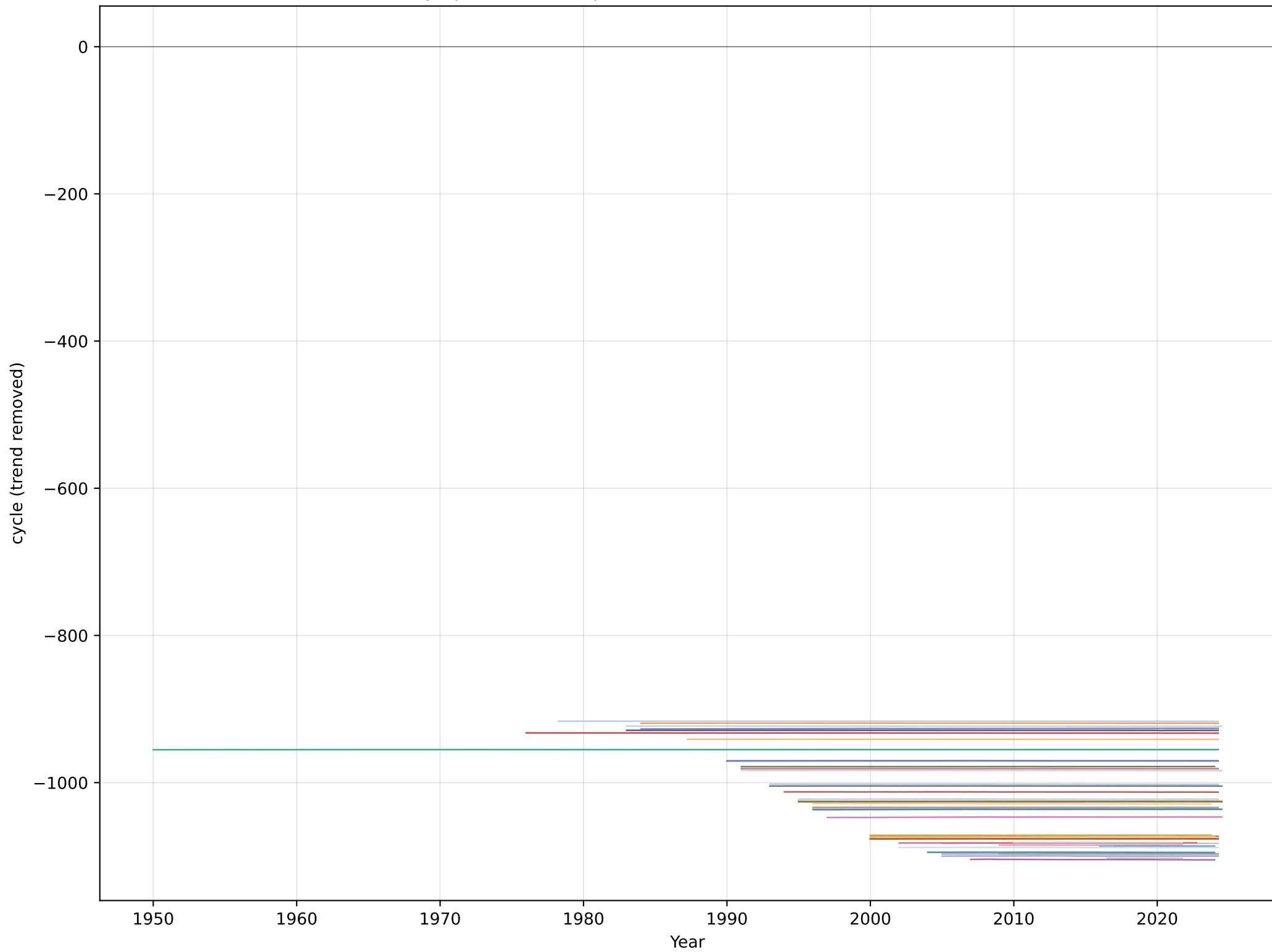
3. Country-specific intercepts, common linear trend

SOE sample: log real GDP per worker countries in file=62 obs=7004 1950-Q1–2024-Q3
Regimes: 3 countries: 62 obs: 7004 log-likelihood: 16544.32
Converged: True CONVERGENCE: RELATIVE REDUCTION OF F <= FACTR*EPSMCH
a_i mean=1024.9814 min=912.4633 max=1102.4355 g 0.0145

Regime AR(1) and transition matrix (SEs in parentheses)

	regime 0	regime 1	regime 2
mu	-1.2116	0.0000	1.2104
		(—)	
sigma	0.8118	0.0175	0.8113
rho	0.8522	0.9801	0.8535
Pi from 0	0.8710	0.0663	0.0626
Pi from 1	0.0260	0.9392	0.0348
Pi from 2	0.0596	0.0634	0.8770

3. Country-specific intercepts, common linear trend — detrended series



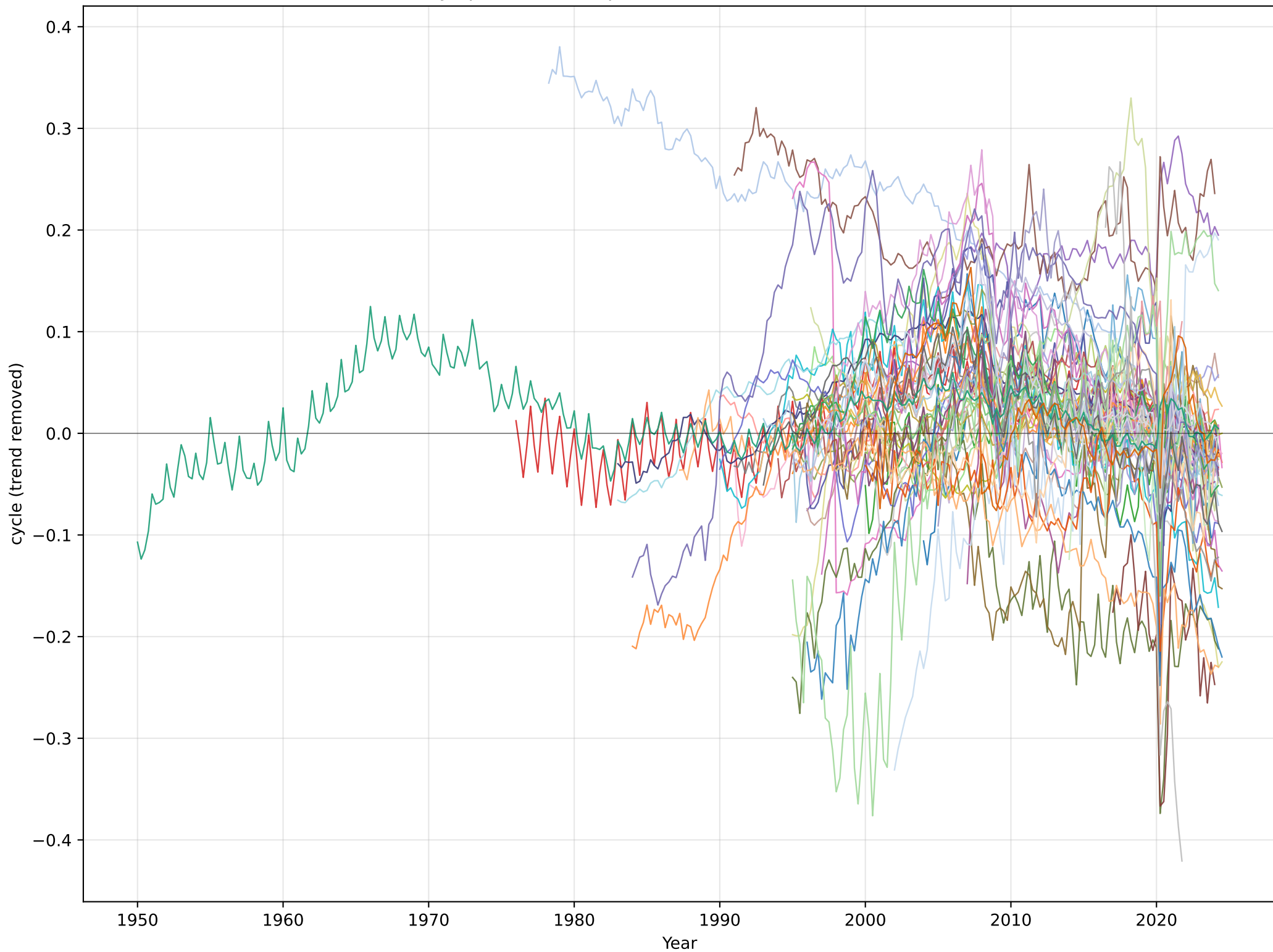
4. Country-specific intercepts and linear trends

SOE sample: log real GDP per worker countries in file=62 obs=7004 1950-Q1–2024-Q3
 Regimes: 3 countries: 62 obs: 7004 log-likelihood: 18395.14
 Converged: True CONVERGENCE: RELATIVE REDUCTION OF F <= FACTR*EPSMCH
 a_i mean=-4.1894 min=-13.0365 max=1.3838 g_i mean=0.0172 min=-0.0318 max=0.1276

Regime AR(1) and transition matrix (SEs in parentheses)

	regime 0	regime 1	regime 2
mu	-0.0181	0.0000	0.0113
	(0.0022)	(—)	(0.0088)
sigma	0.0507	0.0205	0.0092
	(0.0016)		(0.0002)
rho	0.4350	0.9797	0.9596
		(0.0022)	(0.0030)
Pi from 0	0.8282	0.1186	0.0533
	(0.0252)	(0.0264)	(0.0149)
Pi from 1	0.0181	0.9655	0.0164
	(0.0032)	(0.0045)	(0.0033)
Pi from 2	0.0208	0.0077	0.9715
	(0.0035)	(0.0036)	(0.0037)

4. Country-specific intercepts and linear trends — detrended series



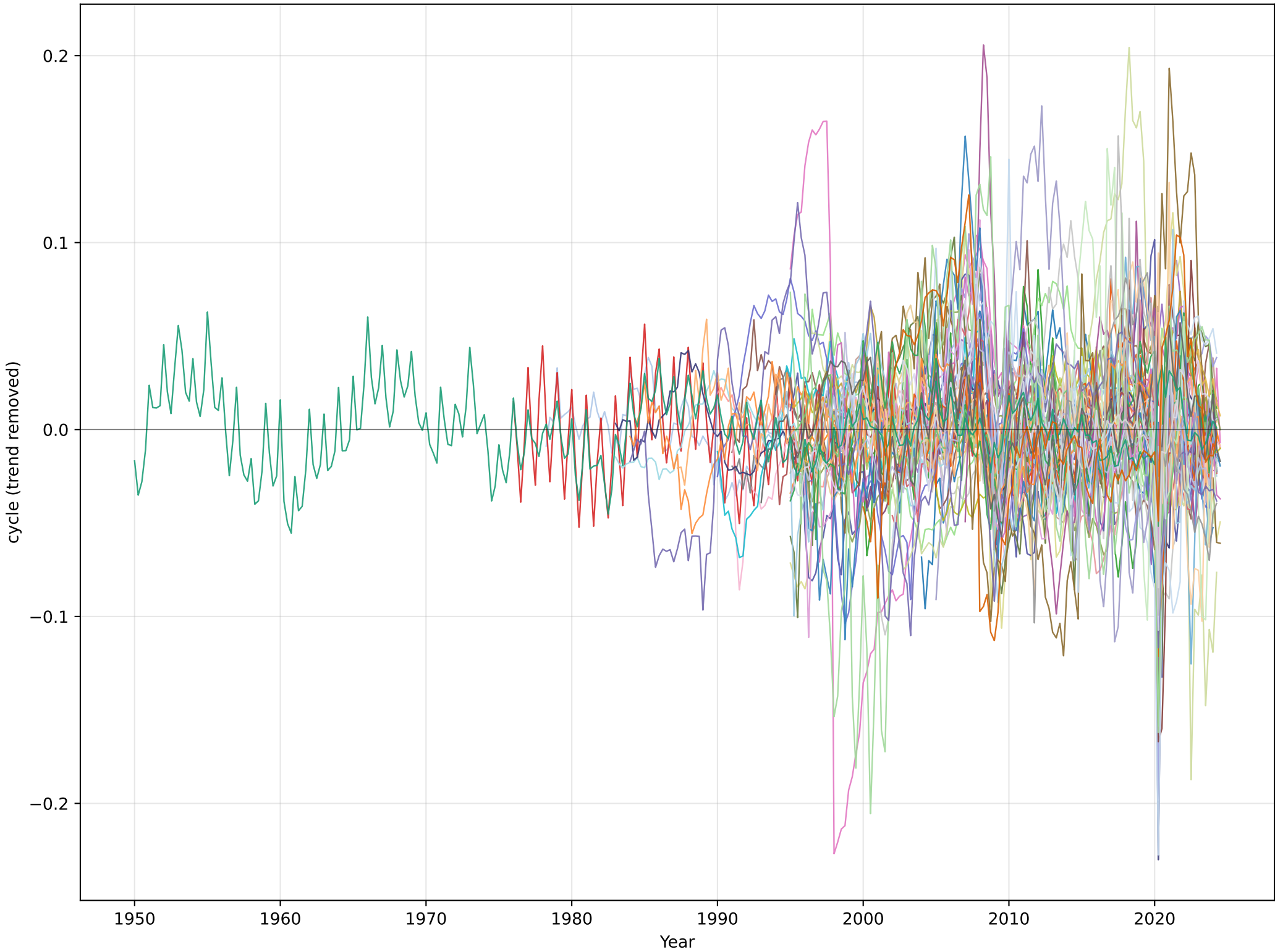
5. Country-specific CF filtered trend (25 years)

SOE sample: log real GDP per worker countries in file=62 obs=7004 1950-Q1–2024-Q3
 Regimes: 3 countries: 62 obs: 7004 log-likelihood: 18637.48
 Converged: True CONVERGENCE: RELATIVE REDUCTION OF F <= FACTR*EPSMCH
 CF low-pass: periods longer than 100 obs (25 years)

Regime AR(1) and transition matrix (SEs in parentheses)

	regime 0	regime 1	regime 2
mu	-0.0173	0.0000	0.0027
	(0.0054)	(—)	(0.0025)
sigma	0.0582	0.0091	0.0204
	(0.0026)	(0.0002)	(0.0005)
rho	0.4857	0.9242	0.8484
	(0.0470)	(0.0072)	(0.0115)
Pi from 0	0.7641	0.0682	0.1676
	(0.0317)	(0.0193)	(0.0325)
Pi from 1	0.0191	0.9725	0.0085
	(0.0033)	(0.0034)	(0.0031)
Pi from 2	0.0208	0.0164	0.9628
	(0.0044)	(0.0040)	(0.0055)

5. Country-specific CF filtered trend (25 years) — detrended series



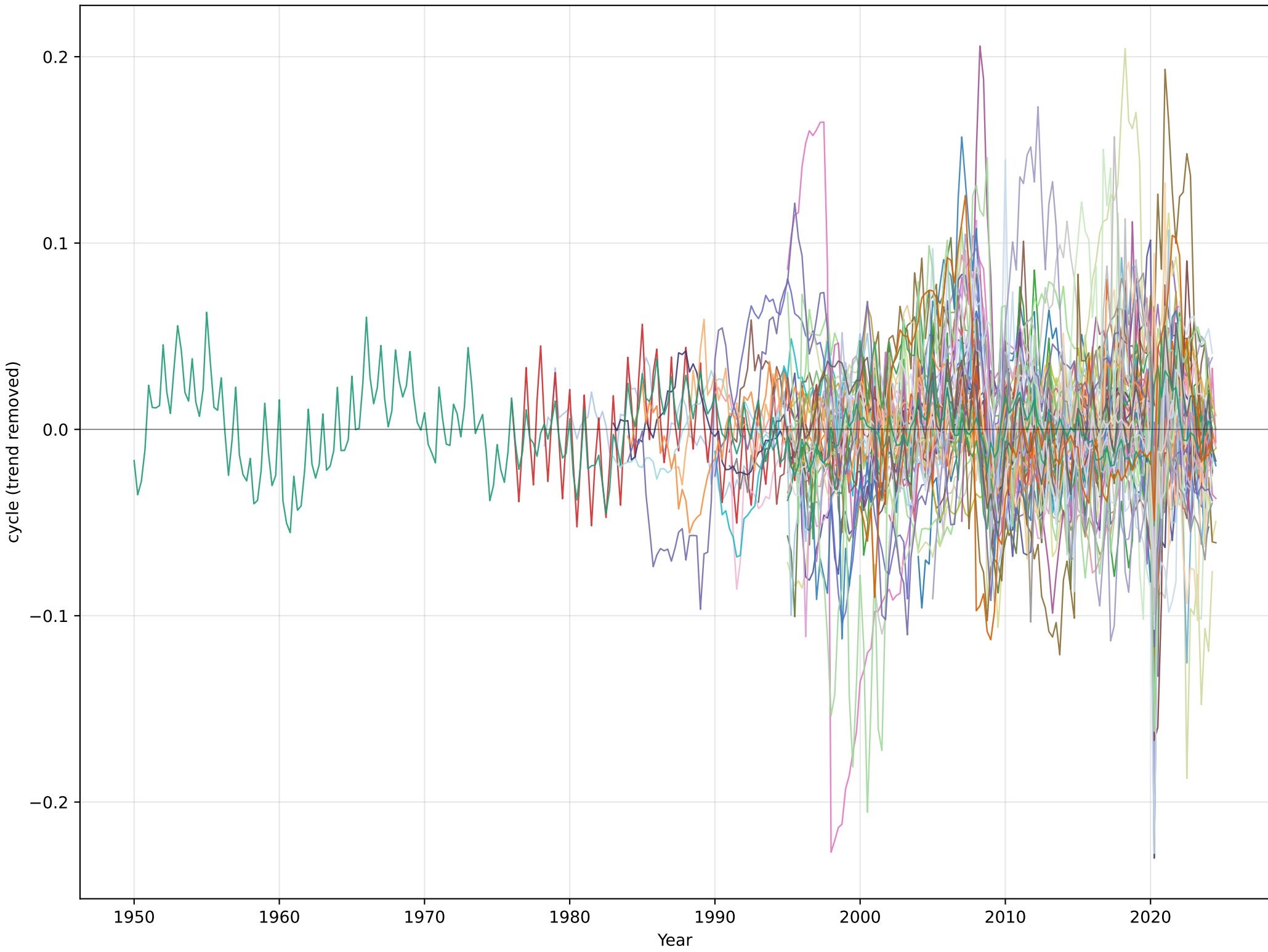
6. CF filtered trend, all regime means restricted to 0

SOE sample: log real GDP per worker countries in file=62 obs=7004 1950-Q1–2024-Q3
 Regimes: 3 countries: 62 obs: 7004 log-likelihood: 18632.52
 Converged: True CONVERGENCE: RELATIVE REDUCTION OF F <= FACTR*EPSMCH
 CF low-pass: periods longer than 100 obs (25 years)

Regime AR(1) and transition matrix (SEs in parentheses)

	regime 0	regime 1	regime 2
mu	0.0000	0.0000	0.0000
	(—)	(—)	(—)
sigma	0.0091	0.0204	0.0589
	(0.0002)	(0.0005)	(0.0027)
rho	0.9264	0.8441	0.5308
	(0.0091)	(0.0155)	(0.0501)
Pi from 0	0.9723	0.0082	0.0194
	(0.0034)	(0.0031)	(0.0034)
Pi from 1	0.0165	0.9638	0.0198
	(0.0040)	(0.0055)	(0.0044)
Pi from 2	0.0681	0.1613	0.7706
	(0.0191)	(0.0325)	(0.0322)

6. CF filtered trend, all regime means restricted to 0 — detrended series



7. CF filtered trend, common rho

SOE sample: log real GDP per worker countries in file=62 obs=7004 1950-Q1–2024-Q3
 Regimes: 3 countries: 62 obs: 7004 log-likelihood: 18572.70
 Converged: True CONVERGENCE: RELATIVE REDUCTION OF F <= FACTR*EPSMCH
 CF low-pass: periods longer than 100 obs (25 years)

Regime AR(1) and transition matrix (SEs in parentheses)

	regime 0	regime 1	regime 2
mu	-0.0304	0.0000	0.0017
	(0.0246)	(—)	(0.0032)
sigma	0.0656	0.0093	0.0208
	(0.0030)	(0.0002)	(0.0005)
rho (common)	0.8822	0.8822	0.8822
	(0.0061)	(0.0061)	(0.0061)
Pi from 0	0.7624	0.0661	0.1715
	(0.0322)	(0.0202)	(0.0340)
Pi from 1	0.0172	0.9742	0.0086
	(0.0030)	(0.0034)	(0.0032)
Pi from 2	0.0191	0.0166	0.9643
	(0.0041)	(0.0041)	(0.0055)

7. CF filtered trend, common rho — detrended series

